DownUnder: Roadmap to a Fundable Product

DownUnder helps women check whether a medication interacts with their birth control or their other medications. This document sets out the problem it solves, how it grows from a product into a platform, and a phased timeline to get there.

The Problem DownUnder Solves

Birth control and a woman's other medications are managed by different people, and no one is reliably responsible for checking how they affect each other. The gap runs in two directions, and both are documented.

Other medicines can weaken birth control, and the rules are hard to remember

Combined hormonal methods (pill, patch, ring) are broken down by the liver's cytochrome P450 enzymes, so a drug that speeds those enzymes up can clear the hormones faster. Cleveland Clinic names seizure medicines (carbamazepine, phenytoin, topiramate), the HIV drug efavirenz, the tuberculosis antibiotic rifampin, and St. John's wort as the main concerns. The same source says most antibiotics and most other medications do not interact, so the real task is separating the few that matter from the many that don't. The effect can also run the other way: combined hormonal birth control can make the seizure medicine lamotrigine less effective.

Medicines that require reliable contraception are often prescribed without it

In a chart review of two family-medicine clinics at one academic center, 25% of women aged 13 to 45 had been prescribed at least one medication with potential risk in pregnancy, and 55% of those women had no documented contraception (JABFM, 2019). Women under 25 had about half the odds of receiving contraception alongside such a prescription (adjusted odds ratio 0.47), and that is the age group campuses serve. The authors also note that a specialist may prescribe such a drug without telling the primary care clinician.

Two cautions apply. It is one center with 2015 data, and ondansetron accounted for 73% of the flagged women. With ondansetron excluded, 62% of the flagged women still had no documented contraception.

The stakes are real, but interactions are one slice of them

In 2011, 45% of US pregnancies were unintended, and the first-year typical-use failure rate for the pill was 7% in 2006 to 2010 data (Guttmacher, 2017). Those figures include every cause. Pregnancies among contraceptive users are almost entirely due to inconsistent or incorrect use (Frost and Darroch, 2008), and none of the sources here measures how many failures come from drug interactions.

So the case for DownUnder is not that interactions explain most unintended pregnancies. It is that this is a specific, preventable failure that no one owns, and that the same profile can support adherence help and wider medication-safety checks later.

Why it is a women's health problem more broadly

A 2020 review of 86 drugs found higher blood concentrations in women for 76 of them, and women experienced adverse drug reactions nearly twice as often as men (Zucker and Prendergast, 2020). The review is about dosing and side effects in general, not contraceptive interactions, but it supports the point that women's medication safety is under-served.

What DownUnder does about it

It checks both directions in one profile: whether a medication may weaken the selected birth control, and how it interacts with the rest of the medication list. It explains the result in plain language, shows its sources, and says "ask a pharmacist" when the data is unclear.

One claim still needs testing: that students are more exposed to this fragmentation because they may get care from a campus clinic, urgent care, telehealth, and home providers. None of the sources above measures students specifically, so the pilot should.

What DownUnder Is Today

DownUnder is a working full-stack MVP: a consumer-style interaction checker built on public data. It covers the whole loop from profile and birth control method to medication search, risk level, plain-language guidance, source links, and disclaimers. Running locally with a single profile is also a real privacy advantage at this stage.

Four gaps matter to buyers:

1. **No predictive layer yet.** A rules-based normalization over public APIs is a lookup tool. The predictive layer is the part still to build.

2. **The interaction data source needs a decision.** NLM discontinued RxNav's drug-interaction features on January 2, 2024, and notes that the sources it drew on, a high-priority interaction list and DrugBank's dataset, may remain available elsewhere (NLM RxNav FAQ). The README uses RxNav for typeahead search, which is a different function. I haven't read the code, so confirm that nothing still calls the retired endpoint and that an empty or failed lookup never displays as "no interaction found." If interactions are inferred from openFDA label text, that parsing is the most fragile part of the app and also the best opportunity.
3. **No accounts or multi-tenancy.** A single local profile is great for a demo, but a university can't buy it in that form.
4. **No clinician-facing view.** Every result so far goes to the student. A pharmacist or campus clinician view matters for the FDA path (see Risks) and for the pilot.

Where AI and Prediction Fit

Use ML where rules don't scale, and keep clinical rules deterministic where published guidance already exists. Eligibility categories such as the CDC's US MEC (smoking, migraine with aura, clotting risk) and interaction guidance from bodies such as the FSRH are clinical standards. They anchor the rules layer, and a model should not override them. Calling the whole product "predictive" would hurt credibility with clinicians, so name each model's job precisely:

- **NLP extraction from label text.** Train or prompt-tune a model to extract and grade contraceptive-relevant statements from openFDA labels (for example "reduces effectiveness," "backup method recommended," "contraindicated"). Every output must point to the label passage it came from so a reviewer can check it, and the rules layer stays as the baseline to beat.
- **Confidence scoring and abstention.** Predict when data is incomplete or sources conflict, and route to "ask a pharmacist" instead of guessing. Safety-minded buyers value this more than extra features.
- **Cohort analytics once there are users.** Aggregate, de-identified patterns such as common medication and method combinations on a campus, and where students get stuck. This is the dashboard a health center would pay for.
- **Adherence and method-switch modeling.** Valuable because most contraceptive failures come from inconsistent use (see the Problem section), but only after real usage data exists. It is planned for Phase 3 below, not promised earlier.

The credibility move is a validation set. Have a pharmacist or clinician label about 150 to 300 pairs, covering both directions (a medication weakens birth control, and a medication needs reliable contraception). Then report how often the rules engine and the model agree with the labels, with precision and recall and special attention to dangerous misses. That one table can anchor a grant application, a pilot proposal, and an investor deck.

From Product to Platform

A product answers one question for one user; a platform lets new questions, users, and partners plug into the same core. DownUnder's first question is whether a medication interacts with someone's birth control. It becomes a platform when the profile, the interaction engine, and the clinician tools are reusable pieces that new modules and partners can build on.

What changes between product and platform

Layer	Today (product)	Target (platform)
Health profile	One local profile with metrics, method, and medications	Consented, longitudinal profile covering medications, methods, conditions, and life stage
Engine	Rules plus label lookups inside the app	Versioned interaction and risk engine with an API, validated against the gold set
Users	A student on her own device	Students, alumni, pharmacists and clinicians, campus administrators
Modules	Birth control and medication check	Adds the teratogen-and-contraception check, hormonal conditions, pregnancy and lactation, menopause, preventive care
Buyers	One campus health center	Adds telehealth, pharmacies, employers, and health plans
Distribution	Direct, campus by campus	Partner API and health-record integrations

Five expansion stages

1. **Contraception and medication safety (the MVP).** Check medications and supplements against the selected method and the rest of the medication list, then add the reverse check for medications that require reliable contraception.
2. **Adjacent hormonal and reproductive conditions.** PCOS, endometriosis, heavy or irregular periods, and hormonal acne, where treatments overlap with contraception and carry interaction questions.
3. **Pregnancy planning and lactation.** Preconception and breastfeeding medication checks, using the labeling and reference sources that already exist for these cases.
4. **Perimenopause and menopause.** Hormone therapy and symptom-treatment decisions carry many of the same interaction and risk-profile questions.
5. **Preventive care and personalized recommendations.** Screening and vaccination reminders (for example cervical screening and HPV vaccination), symptom and cycle tracking, and tailored suggestions about when to talk to a clinician.

How the AI layer scales

- **Longitudinal data creates the moat.** Each added stage deepens the profile over time, which enables risk models, adherence prediction, and personalization that a one-time lookup tool can't match.
- **Use refines the recommendations.** Consented, de-identified patterns across users can show which questions to ask, which warnings matter, and when to route someone to a clinician.
- **Confidence-aware outputs stay central.** The "ask a pharmacist" design that protects users now is what makes broader recommendations responsible later.

How the business scales

- **Retention past graduation.** A student who uses DownUnder in college can keep it afterward, so the product isn't capped at the student years.
- **New buyers at each stage.** Campus health centers first, then telehealth companies, pharmacies, employers and benefits providers, health plans, and women's health clinics.
- **A system of record for a woman's medication and hormonal context.** That is a stronger investor story than a single-purpose app, but only after Stage 1 is proven.

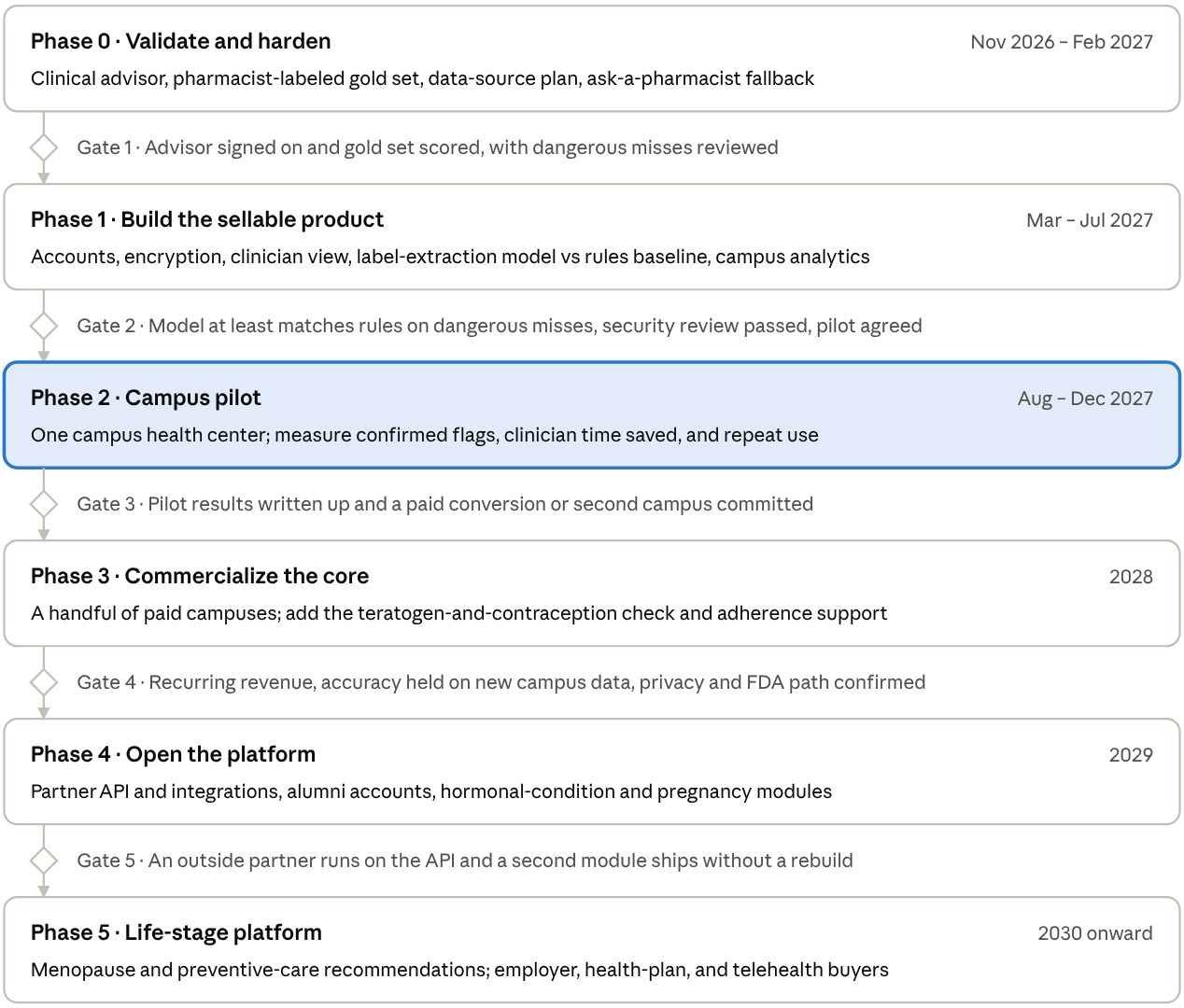
Three signs it has become a platform

1. An outside partner runs on the DownUnder API.
2. A second module ships without rebuilding the core.
3. Clinicians at more than one institution use the same console.

Timeline: Phases and Gates

The plan runs about four years in six phases, and each phase must pass a gate before the next begins. Dates assume a small team working part-time and shift with when a campus agrees to a pilot. Universities buy on academic calendars, so a pilot that misses the August start slips to the next semester.

Each phase passes a gate first; the campus pilot is the proof



roadmap · 6 phases, 5 gates

Phases 0 to 2 prove the narrow product, Phase 3 turns it into revenue, and Phases 4 and 5 open it into a platform only after the pilot gate passes.

The next twelve months in detail

Phases 3 to 5 stay light on detail on purpose, because they depend on what the pilot shows.

Phase 0 (Nov 2026 to Feb 2027)

- Recruit a clinical advisor, ideally a pharmacist or campus health provider.
- Build the labeled gold set and an evaluation script in the repo.

- Decide the interaction data source (license or curate), add caching and fallbacks for the external APIs, and make "unknown, ask a pharmacist" a first-class result.
- Book a regulatory and privacy consult.

Phase 1 (Mar to Jul 2027)

- Add accounts, a hosted database such as Postgres, encryption, and audit logging. SQLite via sql.js is fine for the MVP but not for a campus deployment.
- Add the clinician-facing view.
- Benchmark the NLP extraction model against the rules engine.
- Add de-identified aggregate analytics for administrators.
- Submit a first grant application (NIH SBIR/STTR or a women's-health foundation; check current due dates) and secure a pilot agreement.

Phase 2 (Aug to Dec 2027)

- Run the pilot at one campus health center, ideally one where you already have a contact.
- Track pharmacist-confirmed flags, follow-up advice delivered, clinician time saved, and repeat use.
- Write up the results for grants and for the next campus.

Risks to Settle Early

The biggest risks are regulatory scope, health-data privacy, and claims the evidence can't yet support. This is not legal advice; get counsel before charging for the product.

- **FDA path.** Software is excluded from device regulation as "non-device CDS" only if it meets four criteria, including that it supports recommendations to a healthcare professional who can independently review the basis. FDA's January 2026 revision keeps that framework and announced nothing new for consumer-facing tools, and patient-facing software that meets the device definition stays under FDA's existing digital health policies (Covington summary, FDA guidance page). In practice a clinician-facing view with visible sources is the cleaner route, and a student-facing "this is safe for you" verdict carries more exposure. Present results as information to discuss with a clinician.
- **Privacy and who the customer is.** Contraceptive, medication, and pregnancy-related data is highly sensitive. Which regime applies (HIPAA, FERPA, or state law) depends on the customer and setting, so confirm it with counsel before signing a pilot. Design

for data minimization, strong consent, and de-identified analytics from the start, and keep the local-only design as long as it serves the product.

- **Accuracy and liability.** A wrong "safe" answer could lead to a pregnancy or a harmful combination. Use authoritative sources, keep a pharmacist or clinician reviewing, and treat "unknown, ask a pharmacist" as a valid result.
- **Evidence and claims.** Don't claim fewer unintended pregnancies until they're measured. Pilot outcomes should be process measures (flags reviewed, follow-up advice given, clinician time saved). The statistics in this document date from 2011 to 2020, so refresh them before external use.
- **Scope creep.** Each expansion stage widens the regulatory and privacy surface. Prove Stage 1 first, bring counsel in before each new stage, and keep a clinician in the loop: DownUnder helps women prepare for and use clinical care, and does not replace it.

How to Pitch It

Lead with the problem, then the proof, then the path.

> "Hormonal birth control is common, but the medications that can weaken it, and the medications that require reliable contraception, are prescribed by different providers who may not see the whole list. DownUnder checks both directions in one profile, shows its sources, and tells students and campus clinicians when to ask a pharmacist. We validate its accuracy against clinician-labeled data, prove it in a campus pilot, and then extend the same core into pregnancy, hormonal conditions, and menopause, becoming the medication-safety layer for women's health."

That frames a problem, a customer, a measurable result, a defensible AI component, and a path to a larger market.

Possible Next Deliverables

- Validation-set template and evaluation script
- NLP extraction approach for openFDA labels
- Pilot proposal for a campus health center

Sources

Checked Nov 7, 2026. Pages opened and read:

- NLM RxNav FAQ: drug-interaction features discontinued January 2, 2024
- Sundaram et al., Guttmacher, 2017: 45% unintended pregnancies in 2011; pill failure rate 7%
- Panchal et al., JABFM, 2019: teratogenic prescriptions and contraception in family medicine
- Cleveland Clinic: which medications affect hormonal birth control
- Covington summary of FDA's 2026 CDS guidance

Seen only as abstracts or search summaries, so open them before quoting externally:

- Zucker and Prendergast, 2020: sex differences in pharmacokinetics and adverse drug reactions
- Frost and Darroch, 2008: contraceptive pregnancies mostly due to inconsistent use
- FDA CDS guidance page